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References:

1. A. Matlack, Introduction to green chemistry, CRC Press, 2010
2. V.K. Ahluwalia, Green Chemistry: A Text book, Narosa publishing house, 2020

05/11/2022

Faculty **HOD** **Dean Academics**

Unit-5 Green chemistry

Lecture-1 Definition and Concept of green chemistry

Green Chemistry

✓ The design of processes that reduce or eliminate the use and production of toxic products is known as green chemistry.

✓ The term was first used by Paul T. Anastas in the last decade of the 20^{th} century

✓ Green Chemistry implies:

- Prevention of pollution rather than treatment of pollution
- Environmentally Benign Chemistry
- Sustainable Chemistry
- Ecofriendly Chemistry
- Clean Chemistry

Green chemistry is generally aimed at

➢ Producing chemicals which are safe for biotic as well as abiotic environment.

➢ Using cost and energy effective methods and procedures

➢ Designing processes that reduce or eliminate the use and production of toxic materials

➢ Minimizing the production of wastes.

➢ Avoiding the production of non-biodegradable materials/products.

➢ Maximizing the use of raw-materials from renewable resources

Lecture-2 Principle of green chemistry, waste and pollution prevention hierarchy

Principles of Green Chemistry-

This is true that green chemistry is an alternative tool for reducing pollution. Following principles justify the same:

1. Prevention: Preventing waste is better than cleaning up debris. It helps inefficient use of resources and prevents waste.

2. Atom Economy: Design syntheses so that the final product contains the maximum proportion of the starting materials. Waste few or no atoms.

3. Design less hazardous chemical syntheses: Design syntheses to use and generate substances with little or no toxicity to either humans or the environment.

4. Designing Safer Chemicals: Design chemical products that are fully effective yet have little or no toxicity.

5. Safer Solvents and Auxiliaries: Auxiliary substances should be avoided wherever possible, and as non-hazardous as possible when they must be used.

6. Increase energy efficiency Run chemical reactions at room temperature and pressure whenever possible

Energy processes should be conducted at ambient pressure and temperature whenever possible.

7. Use of Renewable Feedstock's: Renewable feedstocks or raw material should always be preferred to non- renewable whenever technically and economically practicable.

. The source of renewable feedstocks is often agricultural products or the wastes of other processes; the source of depletable feedstocks is often fossil fuels (petroleum, natural gas, or coal) or mining operations.

8. Reduce Derivatives: Avoid using blocking or protecting groups or any temporary modifications if possible. Derivatives use additional reagents and generate waste.

9. Catalysis: Minimize waste by using catalytic reactions. Catalysts are effective in small amounts and can carry out a single reaction many times. They are preferable to stoichiometric reagents, which are used in excess and carry out a reaction only once.

10. Design for Degradation: Chemical products should be designed so that it does not have any harmful impact on the environment. Chemical products should be broken into non- toxic products.

11. Real-time Analysis for Pollution Prevention: Further development of analytical methodologies must be made to allow for real-time, in-process monitoring, and control before the formation of hazardous substances.

12. Minimize the potential for accidents: Design chemicals and their physical forms (solid, liquid, or gas) to minimize the potential for chemical accidents including explosions, fires, and releases to the environment.

Waste and Pollution prevention hierarchy

Pollution prevention is the reduction or elimination of wastes and pollutants at their sources. For all the pollution that is avoided in the first place, there is that much less pollution to manage, treat, dispose of, or clean up.

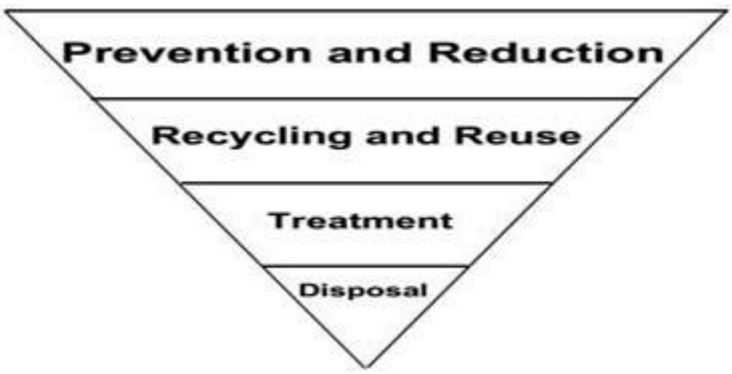
The Pollution Prevention Act of 1990 explains clearly the hierarchy as follow:

i. **Prevention:** it is the most desirable option of the hierarchy and the most effective way to reducerisk. Pollution should be prevented or reduced at the source whenever feasible.

ii. **Recycling:** pollution that cannot be prevented should be recycled in an environmentally safemanner whenever feasible.

iii. **Treatment:** pollution that cannot be prevented or recycled should be treated in anenvironmentally safe manner whenever feasible; and

iv. **Disposal:** disposal or other release into the environment should be employed only as a last resortand should be conducted in an environmentally safe manner.



The waste hierarchy

The waste hierarchy is a simple ranking system used for the different waste management options according to which are the best for the environment.

i. **Prevention**

The idea of avoiding things becoming waste in the first place is essential and the most preferred option in the waste hierarchy. It means not generating waste in the first place or minimising the amount of waste produced.

ii. **Reuse**

When waste is created, the waste hierarchy prioritises reuse. Where possible, reusing waste reuse (for example, refilling a drinks bottle), followed by recycling (processing of wastes into new raw materials).

iii. **Recycling**

Recycling is the most environmentally friendly solution. Recycling turns waste into a new item or product, reducing the amount of raw materials required.

iv. **Recovery**

The recovery of energy by burning or biological treatment. Composting is also a method we use when we can't recycle materials. Composting turns organic wastes into nutrient-rich food for plants.

v. **Disposal**

Land filling is the final option for any wastes that cannot be dealt with in any other way. A landfill site is an area of land set aside for the final disposal of solid waste.

Lecture-3 Green chemistry and sustainability development

Green Chemistry and Sustainable Development

Development that meets the needs of the present without compromising the ability of future generations to meet their own needs.

Green Chemistry and Sustainability

Green chemistry is a tool in achieving sustainability

–Not a solution to all environmental problems

–Fundamental approach to pollution prevention

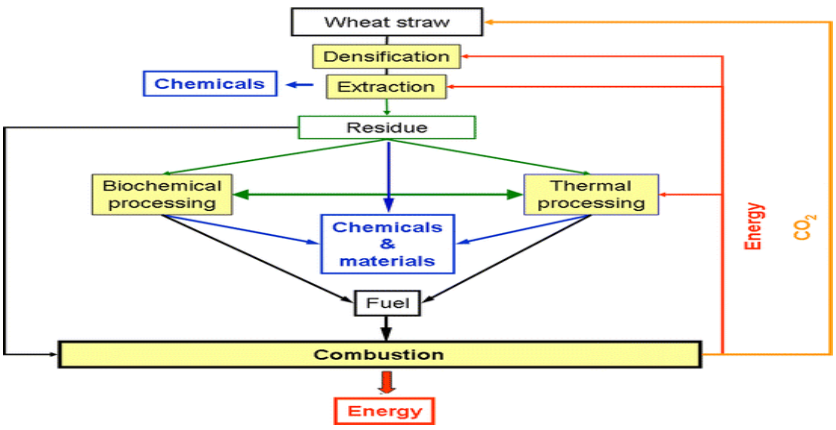
–Chemistry's unique contribution to sustainability

Lecture-4 Use of alternatiove feedstock (bio fuel), green solvent

Biomass as a Chemical Feedstock

Green chemistry aims to develop the greener synthesis of the required chemical products by using the renewable resources(e.g. biomass rather than petrochemical feedstock's). It encourages the use of starting material(i.e. raw material or feedstock)which should be renewable, if technically and economically practicable. In fact, continuous use of the nonrenewable feedstock (e.g. petroleum product, fossil fuel) will deplete the resource and future generation will be deprived. Moreover, use of the nonrenewable resources puts a burden on the environment.

On the other hand, use of sustainable or renewable resources e.g agricultural or biological product ensures the sharing of resources by future generation. Moreover, this practice generally does not put much burden on the environment. The products and wastes are generally biodegradable. The practice of this principle has been illustrated in many cases like bio plastic and biopolymer, biodiesel, CO2 feedstock in the manufacture of polycarbonate, greener synthesis of furfural from biomass, greener synthesis of adipic acid and catechol from biomass e consumption of non-renewable resources (e.g. Crude oil).

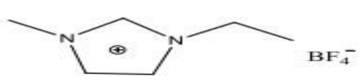


Proposed scheme for an integrated biorefinery

Green solvents- The solvents which are inert, have low toxicity, easy to recycle without contaminating the products are called green solvent. The solvent selected should not have any negative impact on the environment or human health. e.g. halogenated solvents like $CHCl_3$, CCl_4 are suspected carcinogens and to avoid their use, green alternatives like water, ionic liquids, liquid CO_2 have been used..

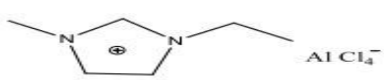
1. H_2O as Green Solvent: Water is regarded as the best solvent for the reactions to be carried out. Water is naturally occurring, non-toxic, non-explosive solvent as against the voc's. However at the same time water is difficult to heat or cool rapidly, its distillation is energy expensive, the contaminated waste streams are difficult to treat.

2. Ionic Liquids An ionic liquid comprises of large nitrogen containing organic cation and a small inorganic anion. Since one part is large and other is small, it creates asymmetry in the compound which makes it a low melting solid. Simple ionic liquids when mixed with the other inorganic salts result in the production of a multi component ionic liquid. Since the components of an ionic liquid are held by strong forces of attraction, it is found that they possess no or low vapour pressure rendering them non-volatile in nature. Further they are non-flammable and non-explosive which is an additional feature rendering them safe to use. They can also be used both as solvent as well as catalysts. Thus the properties of ionic liquids which make them ideal green solvents include: a) Lack of vapour pressure. b) Non flammability and non-explosiveness. c) Stable at high temperature which makes them better for carrying out reactions at high temperatures. d) The property of these ionic liquids can be changed by just changing their concentrations of cations/anions, varying the side chain length in cations. Few of the ionic liquids are mentioned below:



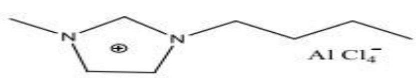
1-ethyl-3-methylimidazolium fluoride-boron (III) triflouride

([bmim] BF_4)



1-ethyl-3-Methylimidazolium chloride-aluminium (III) chloride

([emim]$AlCl_4$)



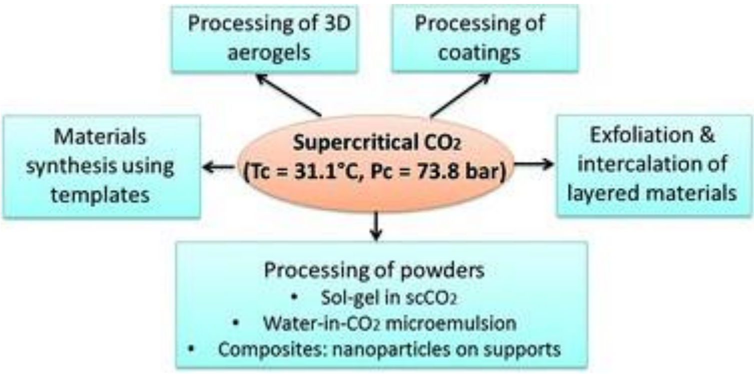
1-butyl-3-methylimidazolium chloride-aluminium (III) chloride

([bmim]$AlCl_4$)

SUPERCRITICAL CARBON DIOXIDE (SCO_2)

It is a fluid state of carbon dioxide where it is held at or above its critical temperature and critical pressure. Supercritical CO2 is becoming an important commercial and industrial solvent due to its role in chemical extraction in addition to its low toxicity and environmental impact. The relatively low temperature of the process and the stability of CO2 also allows most compounds to be extracted with little damage or denaturing. In addition, the solubility of many extracted compounds in CO2 varies with pressure permitting selective extractions.

USES OF SUPERCRITICAL CO2



Lecturer-5 Alternative sources of energy: use of microwaves and ultrasonic energy

Use of microwaves

When the wavelength is in the range 1 mm to 1 m (300 MHz to 300 GHz), is called microwaves. Microwaves are propagated in a vacuum at the speed of light, like any other form of electromagnetic radiation however, they are reflected by metal surfaces. As a result of this property, microwaves can be screened effectively using metal sheets and even close-meshed wire nets. The metal casing and the perforated sheet at the front of household microwave ovens screen the microwaves.

The heating of different materials and substances by the alternating electromagnetic field caused by microwaves results Today, microwave technologies are being tested as energy- and cost-saving alternatives in most areas.

1. **Production of alloys and metal melts**

Metallurgical processes often require high temperatures. This also applies to the production of alloys such as brass and bronze.

2. **Glass production**

The production of glass by simple means is time-consuming. There are formulations which can be melted with a Bunsen burner, but these contain lead compound. With the procedure described here it is possible to produce lead-free glass as well.

Use of ultrasonic energy

Ultrasound is the term used for sound waves in the frequency range 16 kHz to 10 MHz.

The applications of ultrasound is:

1. Ultrasound fields with high frequency (> 2 - 5 MHz) and low intensity are used in diagnostic medicine.
2. Ultrasound methods have been used for decades for testing materials, for example for testing weld seams or for locating hidden cracks in construction parts.
3. Formation of emulsions and the displacement of dirt from surfaces, both for cleaning and activation. For example, ultrasonic baths are used for cleaning purposes.
4. Sensitive surfaces, such as the lenses of glasses, can be cleaned without damaging them in this way. This process also has advantages for deposits that are not easily accessible, but have to be removed. For these reasons, ultrasound baths are used by both opticians and many natural science laboratories.
5. Other applications can be mentioned here: the smashing of kidney stones, the initiation or acceleration of radical reactions, the degradation of organic contaminants in polluted water.